

A Framework for the Education of Radiation Therapists

Endorsed by the African Radiation Oncology Group (AFROG), the Australian Society of Medical Imaging and Radiation Therapy (ASMIRT), the Canadian Association of Medical Radiation Technologists (CAMRT), the European Society for Radiotherapy and Oncology (ESTRO), the Federation of Asian Organizations for Radiation Oncology (FARO), the Ibero-Latin American Association of Radiotherapy (ALATRO) and the International Society of Radiographers and Radiological Technologists (ISRRT)

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FOREWORD

In 2014, the IAEA published IAEA Safety Standards Series No. GSR Part 3, Radiation Protection and Safety of Radiation Sources: International Basic Safety Standards, which establishes requirements for protecting people from harmful effects of exposure to ionizing radiation and for ensuring the safety of radiation sources in all fields of application. The requirements specific to medical exposure include the responsibility of the government to ensure that relevant parties are authorized to assume their roles and responsibilities, as well as the responsibility of the regulatory body to require that professionals involved in the medical uses of ionizing radiation are specialized in the appropriate area and fulfil the requirements for education, training and competence in radiation protection.

Supporting Member States in the safe and effective use of radiotherapy for the treatment of cancer and other diseases is an important component of the work of the IAEA. The successful implementation of radiotherapy depends on the combined efforts of a team of professionals, including radiation oncologists, radiation therapists and medical physicists. Radiation therapists are responsible for the technical aspects of this work, which continue to increase in complexity as the field evolves. Structured academic and clinical training programmes, including radiation protection and safety training, are essential to ensure good practice in this discipline.

Traditionally, education and training in radiotherapy have focused more narrowly on providing the specific knowledge and skills needed to apply particular techniques, which inherently included the ability to perform specific tasks. However, recent technological developments in radiation oncology, together with advancements in tumour biology and personalized treatment strategies, demand a broader set of capabilities to meet the growing challenges. This need can be addressed by providing the knowledge, skills and attitudes that underpin defined competencies in areas such as communication, collaboration, social awareness, organization and leadership. An explicit paradigm known as ‘competency’ has emerged, encompassing the knowledge, skills and attitudes necessary to ensure that professionals are thoroughly and appropriately equipped to meet the societal demands of increasingly complex clinical management strategies. This approach is particularly appropriate for a healthcare discipline such as radiotherapy, in which both robust technical knowledge and practical skills in the workplace are necessary for a safe, high quality practice.

Despite growing recognition that radiotherapy increasingly depends on well trained professionals in the clinical setting, there remains a shortfall of qualified radiation therapists in many countries. This shortage is particularly evident in low and middle income countries, where the profession faces challenges related to professional recognition, job opportunities and appropriate remuneration. Members of regional cooperative agreements and IAEA experts have highlighted the need for support in providing structured academic and clinical training to address these gaps.

Responding to this need, in 2005 the IAEA published Training Course Series No. 25, A Syllabus for the Education and Training of RTTs (Radiation Therapists/Therapy Radiographers), which was replaced by an expanded handbook in 2014: Training Course Series No. 58, A Handbook for the Education of Radiation Therapists (RTTs). Various countries have used these earlier publications as a reference when establishing or upgrading radiation therapist education programmes. However, the rapid development and adoption of the competency based education approach in healthcare disciplines prompted the development of the present framework, which revises and replaces the 2014 handbook.

This publication is intended to support efforts to increase the availability of well trained radiation therapists to an adequate level in countries and regions where there are currently shortages. It also promotes the harmonization of training among educational institutions in IAEA Member States and helps ensure that radiation therapists have access to appropriate education and training to acquire and maintain the necessary qualifications and competencies per international standards and recommendations.

The first draft of this framework was developed in November 2022 during a consultancy meeting of external experts working with IAEA staff. This group included specialists with extensive experience in their respective countries not only in the education of radiotherapy students but also in the preparation of competency based curricula

for training. The IAEA is grateful to all those who contributed to the drafting and review of this publication, in particular N. Anderson (Australia), C. Dickie (Canada), K. Kyei (Ghana), M. Leech (Ireland), E. Rosenblatt (Spain) and G. Wadhawan (India) for their substantial contributions to the drafting process. This framework for the education of radiation therapists has been endorsed by the African Radiation Oncology Group (AFROG), the Australian Society of Medical Imaging and Radiation Therapy (ASMIRT), the Canadian Association of Medical Radiation Technologists (CAMRT), the European Society for Radiotherapy and Oncology (ESTRO), the Federation of Asian Organizations for Radiation Oncology (FARO), the Ibero-Latin American Association of Radiotherapy (ALATRO) and the International Society of Radiographers and Radiological Technologists (ISRRT). The IAEA officer responsible for this publication was S. Ndarukwa of the Division of Human Health.

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1. INTRODUCTION

1.1. BACKGROUND

The need for precision and accuracy in treatment preparation and delivery is a critical component of radiation therapy practice. This necessitates that radiation therapists, the professionals who prepare and deliver radiation therapy courses, have sufficient scientific and clinical expertise, which is only possible through radiation therapy-specific education and training. Irrespective of global location and available technology, the role of the radiation therapist and their level of responsibility are continually evolving and expanding.

Therefore, the profession of radiation therapy, in its own right, needs to be recognized in all jurisdictions as an autonomous expert cancer health professional. To achieve this, national education programmes need to equip the radiation therapist with the knowledge, skills and competences to collate and analyze evidence and synthesize it into the improvement of their own clinical practice, regardless of location and equipment level.

There is a global shortage of well-trained radiation therapists, and this is noted as one of the main barriers to the expansion of radiation oncology services, especially in low- and middle-income countries [1]. In this context, the establishment of education and training programmes is regarded as a high priority by the IAEA [2]. This framework for the education of radiation therapists is a revision of a previous publication: A Handbook for the Education of Radiation Therapists (Training Course Series 58).¹

1.2. OBJECTIVE

The aims of this curricular framework are:

- To provide guidance to education professionals and administrators involved in the education and training of radiation therapists to produce radiation therapy professionals who have the necessary scientific knowledge and are technically and clinically competent.
- To empower radiation therapy educators to significantly increase the radiation therapy-specific content in their education programmes.
- To recognize that the profession of radiation therapist as an expert cancer health professional is necessary for accurate care treatment preparation and delivery.

This publication may also be used as a guide to highlight to policy makers the necessity for radiation therapy education in their countries to improve citizen cancer outcomes.

1.3. SCOPE

All aspects of radiation therapy have been considered in the development of this curricular framework, including the development of a clinical practice education programme which integrates academic learning with hand-on clinical experience in a radiation oncology setting.

¹ (INTERNATIONAL ATOMIC ENERGY AGENCY, A Handbook for the Education of Radiation Therapists, Training Course series 58, IAEA, Vienna (2014).)

Guidance and recommendations provided here in relation to identified good practices represent experts' opinions but are not made on the basis of a consensus of all Member States.

1.4. STRUCTURE

This publication is structured to discuss the education models that underpin radiation therapy. It will then describe the competencies required of a radiation therapist, divided into discrete roles. Logistics issues associated with a radiation therapy education programme are then discussed, followed by a description of teaching, learning and assessment methods that can be utilized to meet the prescribed competencies.

2. SETTING UP AN EDUCATION PROGRAMME FOR RADIATION THERAPISTS

2.1. DEFINITION OF A RADIATION THERAPIST

The radiation therapist is a key member of the radiation oncology team comprised primarily of the clinician (radiation/clinical oncologist), clinically qualified medical physicist and radiation therapist. They are professionals with direct responsibility for the daily administration of radiation therapy to cancer patients. This includes the pre-treatment phase of positioning and immobilization, simulation and in some regions of the world, treatment planning. The radiation therapist is responsible for treatment delivery, including any changes required, as well as the clinical and psychosocial care of the patient daily during treatment and in the immediate post treatment phases. The role of the radiation therapist always encompasses the safe and accurate delivery of the prescribed radiation dose. As the professional in daily contact with the patient, it also includes monitoring, recognizing and reporting side effects as well as providing referral as appropriate with the wider multidisciplinary team.

Within the multidisciplinary team, the professional responsible for the direct administration of the course of radiation therapy is the medical radiation technologist or radiation therapist. Internationally many countries have adopted the title of ‘radiation therapist’. Therefore, The IAEA recommends the global use of the title radiation therapist for this group of professionals.

2.2. EDUCATION OF THE RADIATION THERAPIST

The IAEA is committed to supporting Member States in the process of planning and implementing national education programmes for radiation therapists and has successfully accomplished this in several Member States.

When developing any education programme, it is first necessary to have a complete overview of the current status of radiation therapist education in the country/region, together with the academic and clinical infrastructure already available. This information will assist in defining national needs and ensure that the planning of any new programmes or expansion of existing programmes is cost effective, efficient and effective at meeting the aims of the programme.

Programme planning needs to be outcome-based, meeting local and national workforce planning requirements as per the national cancer control plan, where available. The programme has to encompass a well-defined clinical practice education programme with defined learning outcomes and associated teaching, learning and assessment strategies.

Several approaches to programme design are possible depending on the level of resources, commitment and support available to the programme development working group. However, the overarching principle is that the programmes contain radiation therapy-specific education.

2.3. EDUCATION MODELS

It has long been recognized that learning takes place at an increasing level of complexity from the simple recall of facts to the process of analysis and evaluation. This ascending order of complexity was first described by Benjamin Bloom, a US educationalist (1913–1999) [4] and is still the most widely used taxonomy or classification of the levels of thinking during the learning process. Bloom devised his taxonomy to classify forms and levels of learning in 1956. It was based on the premise that you cannot apply or evaluate until you understand and that

learning at a higher level is dependent on having acquired the prerequisite knowledge and skills at lower levels. In 2001, it was revised by Anderson and Krathwohl [5] to reflect more current approaches to teaching and learning (Fig. 1) but the basic premise remains the same. Bloom believed that teachers should design lessons and tasks to help learners meet the stated objectives, a belief that is still applicable and relevant today, but is now referred to more commonly in terms of aims and learning outcomes.

Bloom identified three domains of learning: cognitive, affective and psychomotor, each with an ascending order of complexity. The levels of complexity achievable are defined in each domain. According to Bloom, the use of the correct verb is the key to the successful writing of learning outcomes.

Learning outcomes have to be written to address the overall programme, individual module and individual lesson levels. Learning outcomes define what the student should be able to do at the end of the unit of learning (be that a single class, a full module or an entire programme). Teaching and learning strategies, as well as assessment strategies, need to be aligned with learning outcomes.

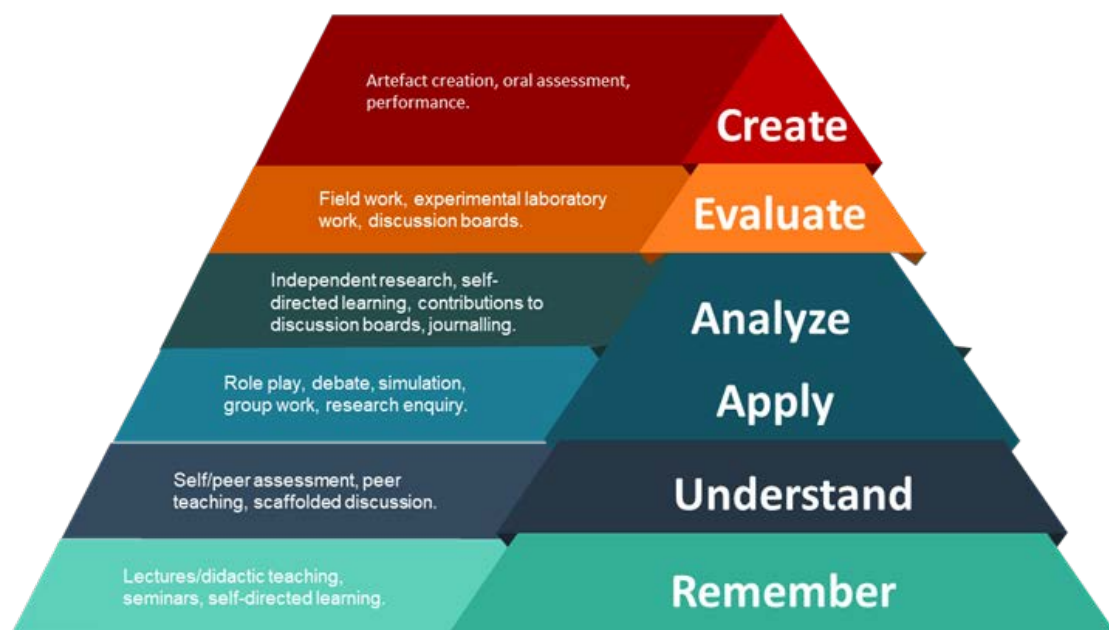


FIG. 1. Bloom's Taxonomy. From Vanderbilt University Center for Teaching. (<https://commons.wikimedia.org/wiki/File:Blooms-Taxonomy-650x366.jpg>), added teaching and learning labels by Academic Practice, Trinity College Dublin, Ireland. <https://creativecommons.org/licenses/by-sa/4.0/legalcode>. (Anderson and Krathwohl) [from: Creative Collaborative Lifelong Learners].

- **Remembering** is retrieving information from long term memory. This could be recall of basic theories and facts of radiation therapy practice and knowledge of the basic sciences. This is linked with teaching strategies including the traditional lecture and other didactic methods, such as seminars.
- **Understanding** is constructing meaning from instructional messages including oral, written and graphic communication. It can be taught using scaffolded (planned, unit-based) discussion and peer teaching and is useful in teaching of radiation therapy techniques.
- **Applying** is carrying out a procedure in a given situation. Teaching strategies that require students to apply their knowledge in a given situation include role play, which is very useful in teaching and assessing student ability to communicate with patients, carers and within the radiation oncology team.
- **Analyzing** is breaking the material into its constituent parts and determining how the parts relate to one another and to the overall structure or purpose. Journaling is an excellent example of a teaching and learning strategy for the development of critical analysis skills in radiation therapy. It is also a useful assessment strategy when students prepare a journal based on their clinical education programme and evidence analysis of their own experience compared to evidence-based practice.
- **Evaluating** is making judgements based on criteria and standards. In radiation therapy education, teaching evaluation of treatment plans is one such strategy to enhance evaluation skills.
- **Creating** is putting elements together to form a coherent whole function: reorganizing elements into new patterns of structure. Creation of independent research and the incorporation of research skills should be an aim of all radiation therapy education programmes.

2.4. CORE COMPETENCIES AND SCOPE OF PRACTICE

It is intended that education institutions within individual Member States develop and use the teaching, assessment and evaluation methods that are most suitable for their situation. This IAEA handbook outlines different useful teaching, assessment and evaluation methods but it is at the discretion of the education institutions within individual Member States as to which methods should be used, how they should be used and when they should be implemented. This should be in accordance with the national guidelines established by the different bodies responsible for teaching programmes and for the certification. However, the programme should focus on radiation therapy-specific content.

There is a need within any curriculum to identify and define both the core competencies that are programme and profession specific, and the generic competencies that are essential workplace skills that cut across occupational and academic titles. Generic competencies are not taught in lessons as such but are fundamental to the philosophy of educational programmes and essential in professional programmes. These generic competencies (core abilities) include: working productively, learning effectively, communicating clearly, working cooperatively, acting responsibly, valuing self positively and thinking critically and creatively. Generic competencies need to be integral to both the academic and clinical programmes. When designing the curriculum, the core competencies can be described in terms of learning outcomes thus defining the appropriate content. They can be very effectively measured as a part of the clinical placement evaluation process.

When defining the content and level of an outcome-based education programme it is important to first define the scope of practice of the graduate radiation therapist. One of the major changes is the shift from a focus based on theoretical knowledge and skills to competency-based education and training. Optimal education/training requires that the student can integrate knowledge, skills and attitudes to perform a professional act adequately in each situation.

The Canadian Medical Education Directives for Specialists (CanMEDS) (Canadian Medical Education Directives for Specialists) system [6] developed by the Royal College of Physicians and Surgeons of Canada framework is one of the most widely adopted medical competency frameworks worldwide [7]. Since its inception and publication, many health professions outside of medicine have elected to adopt this competency framework. At the time of this publication, both Europe and Canada have adopted the Canadian Medical Education Directives for Specialists (CanMEDS) framework as a basis for the competency profiles for the profession of radiation oncology in their jurisdictions. The Canadian Medical Education Directives for Specialists (CanMEDS) framework groups required competencies thematically under distinct roles identified for the profession. Framing competency through the lens of the role radiation therapists play within the healthcare system works to support professional identity formation as well as providing alignment with other key members of the interprofessional radiation therapy team, locally, nationally and internationally.

Current technological developments in oncology require experts who can meet the associated growing challenges. This can be achieved by providing the knowledge skills and attitudes which underpin defined roles including communicator, collaborator and quality care provider.

The transition to competency-based education and training is facilitated by insight into the structure of the learning process (e.g. from ‘theory to practice’) which has a long history in the science of pedagogy, in the past more directed to learning processes in childhood and adolescence, but more increasingly transposed to adults (andragogy). These analyses of major elements of learning (e.g. the internationally well recognized ‘Blooms taxonomy’ previously described) form the basis of the competency-based approach presented here.

Following completion of an education programme at degree level, based on the recommendations in this publication, the radiation therapist needs to have the knowledge, skills and judgement for the competencies outlined under the following seven radiation therapist roles:

- Radiation Therapist Expert;
- Communicator;
- Collaborator;
- Leader and Advocate;
- Quality Care Provider;
- Evidence Based Practitioner;
- Professional.

The roles are described as follows:

Radiation Therapist Expert

The radiation therapist needs to be able to understand and interpret the treatment prescription to accurately prepare and deliver a course of treatment to an individual patient. This requires

knowledge of human anatomy, physiology, cancer as a multitude of diseases, the technology used in radiation therapy and the interaction between them, knowledge of radiation physics, familiarity with radiation therapy doses and schedules and applied mathematics and computing.

The radiation therapist is responsible for simulation and preparation of positioning, immobilization and additional devices as required. As part of this role the radiation therapist will take responsibility for evaluating the treatment intent and interpreting the information given with respect to the patient's disease and general condition to define the most appropriate position and effective immobilization method for set-up reproducibility. In addition, all radiation therapists will be required to understand and evaluate treatment plans and different parameters related to those plans, however their role in treatment plan construction will vary according to the norms of the country and clinical department.

The radiation therapist is responsible for the accurate delivery of the radiation therapy treatment. This includes the actual treatment, accurate maintenance of all daily records, updating of relevant information and careful monitoring of the patient status to detect any physical or psychological changes that require appropriate action. This includes referrals to relevant professionals within the multidisciplinary team and ensuring that all pertinent information is transferred accurately, concisely and in a timely manner.

Verification of the daily set-up is a crucial component of treatment, and the radiation therapist has to be competent to make decisions with respect to the accuracy of the treatment and to adjust within agreed protocols.

Communicator

As the radiation therapist is the person who interacts with the patient daily, they need to always respect the dignity and privacy of the patient. Inter and intra professional communication between all members of the multidisciplinary team is crucial to safe practice.

As a member of the multidisciplinary team, the radiation therapist has to ensure that relevant information is effectively communicated in a timely fashion to the appropriate professionals as well as to patients and their care givers.

The radiation therapist has to understand the radiation therapy process and provide information to staff, patients and their families and members of the public as appropriate and be able to differentiate and fulfil the informational requirements of each group. As the main daily contact point for the patient on treatment, they have to ensure that they educate the patient of the procedures at every stage, answer their questions to increase understanding and continuously monitor the physical and psychosocial status of the patient prior to any procedures taking place.

Collaborator

The radiation therapist is an essential member of the multidisciplinary team that provides optimal care to patients. They understand their role and responsibilities and that of all healthcare team members and work effectively and collaboratively with all. The radiation therapist is respectful and supportive of other healthcare professionals and ensures coordination of care and proper transfer of information and accountability at every care transition.

Leader and Advocate

The radiation therapist is a leader in the radiation medicine environment. They advocate for the radiation therapy profession as well as for patients and their caregivers. The radiation therapist takes the initiative to provide mentorship of colleagues and finds opportunities for leading an effective team to facilitate change management in an evolving healthcare setting. Radiation therapists are stewards of high-quality compassionate care and understand when to reach out for consultation beyond the boundaries of their professional practice.

Quality Care Provider

The radiation therapist, in collaboration with the radiation protection officer and radiation oncology team, ensures the radiation protection and health and safety of staff, patients and the members of the public during the time that they are present within their working area. This includes radiation protection for external beam radiation therapy and brachytherapy procedures as appropriate. The radiation therapist is a reflective practitioner who contributes to the optimization of radiation protection and safety in medical exposures including quality assurance and quality control activities, radiation protection/safety audits, risk management, incident reporting and learning system and prevention and investigation of radiotherapy accidents. The foundation of the practice of the radiation therapist is a commitment to high quality patient care and safety, ethical principles, professional integrity as well as their extensive knowledge of medical and legal requirements.

Evidence Based Practitioner

The radiation therapist engages in the continuous maintenance and development of competence, striving to reach higher levels of expertise and proficiency through integration of evidence-based practice and generation of evidence in practice. They have a responsibility to constantly evaluate and improve practice. As a clinician, the radiation therapist has to address practice gaps, identify opportunities for improvement of processes and procedures, assess technological developments and continuously scan the horizon for emerging issues. With the evidence they acquire from these activities, the radiation therapist disseminates and translates findings into informed practice. For these reasons, the radiation therapist is a lifelong learner and an educator for others, continually building the global body of knowledge.

Professional

The public perception of the radiation therapy department can be directly influenced by the high visibility of the radiation therapist who has to be conscious of always projecting a professional image. This includes professional appearance and manner, self-awareness and self-management, competency limitations, a high standard of ethical and moral behaviour, reliability and responsibility, autonomy and respect for patients, their caregivers and the healthcare team. The radiation therapist has to be insightful of the dynamics of others around them. This includes managing conflicts, practicing and negotiating social awareness, ensuring equity, diversity and inclusive principles are always adhered to.

Table 1 presents a detailed description of the seven roles, as well as key competences and enabling competence(s) of each role.

TABLE 1. SEVEN ROLES, KEY COMPETENCES AND ENABLING COMPETENCES

Roles	‘Key Competence(s)’ ^a	‘Enabling competence(s)’ ^b
1. Radiation Therapist Expert	1.1 Evaluate decision for radiation therapy and brachytherapy	1.1.1 Identify the area for treatment. 1.1.2 Evaluate the patient condition and the limitations that may result from any co-morbid conditions.
	1.2 Decide the optimal position for the patient according to tumour site	1.2.1 Identify the co-morbidities that will impact on patient position. 1.2.2 Ensure consistency, patient comfort and reproducibility of treatment position.
	1.3 Select and/or construct the most appropriate device for the individual patient within the context of the protocol	1.3.1 Apply knowledge of positioning and immobilization to select the appropriate device
	1.4 Acquire simulation images for planning purposes	1.4.1 Apply principles of human anatomy to determine scan borders. 1.4.2 Select the most appropriate imaging modality/ies relevant to tumour site/intent, inclusive of motion management strategy if applicable. 1.4.3 Administer contrast as per local protocol. 1.4.4 Place fiducials as appropriate to site 1.4.5 Apply AS LOW AS REASONABLY ACHIEVABLE (ALARA) principles when selecting optimal imaging parameters.
	1.5 Appraise acquired images prior to data transfer	1.5.1 Confirm images are fit for purpose prior to transfer.
	1.6 Interpret all planning parameters for the clinical site/s and the prescription	1.6.1 Apply knowledge of radiation physics when evaluating the plan.
	1.7 Appraise planning techniques for the clinical site/s	1.7.1 Apply knowledge of fractionation schedules when evaluating the plan. 1.7.2 Assess target volume coverage and organ at risk sparing through the understanding of INTERNATIONAL COMMISSION ON RADIATION UNITS AND MEASUREMENTS (ICRU) recommendations
	1.8 Set-up and treat the patient	1.8.1 Confirm readiness for treatment including consent and patient identification and pre-treatment preparation as per local protocol. 1.8.2 Position and immobilize the patient. 1.8.3 Reproduce planned patient treatment position differentiating localization of reference points and isocenter. 1.8.4 Perform in-vivo dosimetry if appropriate. 1.8.5 Select and acquire appropriate imaging for procedure as per protocol optimizing imaging parameters.

TABLE 1. SEVEN ROLES, KEY COMPETENCES AND ENABLING COMPETENCES (CONT.)

Roles	‘Key Competence(s)’ ^a	‘Enabling competence(s)’ ^b
	1.9 Make treatment decision based on image evaluation	1.9.1 Evaluate acquired image for patient position, isocenter location and, if applicable, volume match
	1.10 Manage the delivery of prescribed radiation therapy	1.10.1 Apply motion management procedure, if applicable 1.10.2 Perform final checks prior to beam-on including time-out or checklists. 1.10.3 Monitor patients during treatment and respond appropriately.
2. Communicator	2.1 Communicate clearly and empathetically with patients and carers	2.1.1 Use effective communication strategies to facilitate patient-informed decision-making. 2.1.2 Communicate effectively, adhering to principles of equity, accessibility, diversity and inclusion. 2.1.3 Provide information personalized to individual circumstances cognizant of patient emotional status, health literacy. 2.1.4 Verify patient understanding. 2.1.5 Provide access to language services as needed. 2.1.6 Adapt communication styles including both verbal and non-verbal communication to the given context.
	2.2 Establish and maintain effective communication with staff and colleagues	2.2.1 Demonstrate self-awareness of own personality traits and communication styles. 2.2.2 Communicate effectively in line with principles of equity, accessibility, diversity and inclusion. 2.2.3 Adapt communication styles including both verbal and non-verbal communication to the given context. 2.2.4 Maintain effective communication with colleagues within and outside the radiation therapy team.
	2.3 Perform documentation and record keeping	2.3.1 Produce clear and objective written communications and reports. 2.3.2 Abide by relevant legislative requirements on appropriate documentation and record keeping.
3. Collaborator	3.1 Collaborate with colleagues within multidisciplinary, research and education teams throughout the radiation treatment pathway	3.1.1 Demonstrate self-awareness of your own competency levels. 3.1.2 Support peer and student well-being and development 3.1.3 Participate in research activities
	3.2 Understand, roles and responsibilities to work effectively with all members of the care team	3.2.1 Respect the context of your role and that of your peers within the multidisciplinary team. 3.2.2 Demonstrate awareness of power dynamics in team interactions. 3.2.3 Use effective interpersonal skills.

TABLE 1. SEVEN ROLES, KEY COMPETENCES AND ENABLING COMPETENCES (CONT.)

Roles	‘Key Competence(s)’ ^a	‘Enabling competence(s)’ ^b
	3.3 Work in partnership with patient and carers	3.3.1 Establish trust with patients and carers. 3.3.2 Demonstrate awareness of power dynamics in patient interactions. 3.3.3 Act in the best interest of patients and carers 3.3.4 Respect patient autonomy and decision-making
4. Leader	4.1 Act as a leader and mentor within the multidisciplinary team	4.1.1 Reflect on the value of mentor-mentee relationship within your specialty. 4.1.2 Respond professionally to constructive feedback. 4.1.3 Foster professional growth of self and peers
	4.2 Advocate for change management	4.2.1 Contribute to change management activities. 4.2.2 Respond professionally to change implementation.
	4.3 Approach conflict productively	4.3.1 Demonstrate awareness of different conflict styles. 4.3.2 Respect the positive attributes of conflict in idea generation. 4.3.3 Encourage respectful conflict resolution.
	4.4 Promote awareness of professional identity of radiation therapists	4.4.1 Engage in activities to promote and support the profession of radiation therapy. 4.4.2 Support peers and students to be advocates of their profession.
	4.5 Advocate for patients and support navigation of healthcare systems	4.5.1 Identify risks and barriers to optimal access to radiation therapy. 4.5.2 Organize delivery of care in a patient-centered manner.
	4.6 Develop strategies to mitigate identified risks and barriers	4.6.1 Facilitate information exchange between patient, carers and the relevant members of the team.
5. Quality Care Provider	5.1 Act within the legal boundaries of the National Radiation Protection and safety legislation	5.1.1 Demonstrate in-depth knowledge of local radiation protection procedures and protocols. 5.1.2 Inspect the area to ensure that radiation protection and safety measures are in place and functional and act or report where necessary. 5.1.3 Ensure radiation protection of staff, students and patients within working environment. 5.1.4 Continuously evaluate and optimize radiation protect strategies within the multidisciplinary team to minimize exposure in accordance with ALARA principles
	5.2 Implement safe work, principles and practices in the department	5.2.1 Inspect the area to identify any health and safety hazards and respond. 5.2.2 Practice safe ergonomic principles 5.2.3 Use personal protective equipment appropriately. 5.2.4 Respond to emergencies involving patients, staff, equipment facility or environment. 5.2.5 Practice infection prevention and control 5.2.6 Perform Basic Life Support

TABLE 1. SEVEN ROLES, KEY COMPETENCES AND ENABLING COMPETENCES (CONT.)

Roles	'Key Competence(s) ^a '	'Enabling competence(s) ^b '
	5.3 Communicate the correct, relevant and appropriate information while adhering to risk management principles in service provisions	5.3.1 Transfer information to allow for correct treatment according to treatment plan and prescription. 5.3.2 Perform transfer of care and accountability to other healthcare providers.
	5.4 Report incidents and near misses	5.4.1 Actively contribute to a no-blame safety culture. 5.4.2 Use existing incident reporting protocols (Local, regional or international). 5.4.3 Examine and learn from any incidents or near misses and how they can be prevented in the future. 5.4.4 Consider risk hazard analysis in implementing new technologies/techniques/processes 5.4.5 Analyse incident reporting over time for trends in safety risk and address 5.4.6 Participate in proactive risk assessment and mitigation strategies to prevent radiation safety incidents in accordance with departmental and regulatory guideline
	5.5 Adhere to local/national quality assurance protocols, processes and standard operating procedures	5.5.1 Perform QA protocols in your institution. 5.5.2 Participate in QA activities 5.5.3 Perform final checks prior to beam-on including time-out and/or checklists. 5.5.4 Participate in regular and independent audits of the QA programme for medical exposures. 5.5.5. Participate in continuous improvement of QA protocols and benchmarking against best practices
	5.6 Adhere to local/national quality control procedures/ protocols	5.6.1 Perform the daily/weekly/monthly quality control for patient procedures, processes and equipment. 5.6.2 Analyse and record the results and report any deviations. 5.6.3 Participate in the investigation of radiation-related incidents and accidents in accordance with departmental and regulatory guidelines.
	5.7 Support and care for the patient holistically before, during and after radiation therapy treatment	5.7.1 Educate the patient as to the treatment procedure and anticipated side effects and their management, prior to treatment. 5.7.2 Assess the daily health status of the patient prior to treatment. 5.7.3 Provide a safe environment for private discussion. 5.7.4 Advise the patient on management of side effects during treatment. 5.7.5 Identify/record all side effects and any intervention recommended. 5.7.6 Refer the patient within the multidisciplinary team as appropriate. 5.7.7 Advise patient on immediate post treatment care. 5.7.8 Inform patient of the follow up procedures.

TABLE 1. SEVEN ROLES, KEY COMPETENCES AND ENABLING COMPETENCES (CONT.)

Roles	‘Key Competence(s)’ ^a	‘Enabling competence(s)’ ^b
6. Evidence Based Practitioner	6.1 Drive evidence generation within the radiation therapist profession	6.1.1 Evaluate research with respect to current departmental practice. 6.1.2 Reflect upon and evaluate your practice regularly. 6.1.3 Identify and critique the appropriate literature in the areas of research, development, innovation, education and clinical practice.
	6.2 Produce evidence to support best clinical practice within the profession	6.2.1 Initiate topics for radiation therapy research and collect outcomes. 6.2.2 Facilitate/participate in local or international clinical trials within own context. 6.2.3 Demonstrate understanding of evolving technologies/innovative treatment options and their impact on radiation therapy delivery.
7. Professional	7.1 Behave according to professional, ethical and legal requirements	7.1.1 Practice in accordance with legislation regulations and ethical guidelines 7.1.2 Demonstrate professional appearance according to local protocols. 7.1.3 Take accountability for your professional practice. 7.1.4 Ensure personal value systems do not impact on professional decision-making. 7.1.5 Maintain privacy and confidentiality. 7.1.6 Ensure ongoing informed consent to procedures. 7.1.7 Treat patients and colleagues with dignity and respect
	7.2 Recognize competency limits	7.2.1 Work within your defined scope of practice and seek help when needed. 7.2.2 Practice within the limits of your professional knowledge, skills and judgement 7.2.3 Seek assistance in instances of insufficient knowledge or skills.
	7.3 Demonstrate care of own well-being to prevent burn-out	7.3.1 Maintain physical and mental health. 7.3.2 Manage personal and professional demands.
	7.4 Promote life-long learning	7.4.1 Participate in continuing professional development. 7.4.2 Take ownership of own professional development. 7.4.3 Reflect critically on own

^aKey competence(s): The knowledge, skills and judgement required of a radiation therapist at entry to practice.

^bEnabling Competence(s): More detailed deliverables of the key competency. Describes the knowledge and skills that contribute to the successful outcomes or delivery of the key competency.

2.5. ENTRY REQUIREMENTS

The universal concept known as ‘secondary or high school education’ implies a level of achievement necessary for preparing the scholars to take their place in adult society. It is recommended, therefore, that the students entering the radiation therapy programme need to have completed the recognized secondary/high school studies which would provide the

foundation for and prepare them for higher education studies [8]. The content of the radiation therapy programme is such that a secondary/high school final diploma including mathematics, life sciences and physical sciences is recommended.

2.6. PROGRAMME DURATION

A minimum 3-year degree programme is necessary to enable the development of the radiation therapist as a key member of the multidisciplinary team and to allow for knowledge and competencies in advanced radiation therapy techniques and standard planning/dosimetry. Where appropriate, a 4-year degree programme could be offered to enable the radiation therapist to participate in advanced planning/dosimetry as well as comprehensive QA. All degree programmes have to also enhance research skills and the development of a professional who can contribute to the advancement of radiation oncology in the country/region.

It is important to remember that an entry level professional education programme is not an end to a learning journey. Developments in radiation therapy are on-going and a well-structured programme of postgraduate education is necessary to enhance clinical practice and widen professional knowledge for all radiation therapists. Continuing professional development or continuing medical education is compulsory in many countries and is seen as an integral part of being a professional. Experience gained in professional practice can provide part of this on-going education provided such experience is recorded and supported by participation in academic programmes [9–12].

2.7. EDUCATION LEVEL AND CERTIFICATION

It is strongly recommended that the radiation therapy programme is offered through a collaborative partnership of a Higher Education Institution (University or College) and one or more clinical sites.

Although there is wide variation in education programmes, radiation therapy qualifications are already well established in many low-, middle- and high-income countries. This framework has been produced to aid radiation therapists in achieving degree level standard education as the entry requirement into the profession.

It is important that qualifications gained by radiation therapists as part of a formal education programme be certified by a competent authority or University. The degree has to be issued by a competent national authority confirming that an education programme is recognized and the student has successfully completed it.

It is desirable that any qualification is defined under an internationally recognized qualifications framework operational in the corresponding region. Such frameworks are a useful mechanism to define the level of learning achieved by the student. The aim of defining a qualification in this way is to provide international transparency, recognition leading to an equality of education and ease of mobility of personnel between countries. These frameworks or national mechanisms do not prescribe content but provide a context for the design of new qualifications consistent with the aims of this publication. Furthermore, a framework can act to bring about change and can be used to indicate where new programmes are required to meet societal needs including the introduction of new qualifications.

These qualification frameworks assist in the development of education programmes at several levels, facilitating institutions of education to prepare new programmes or extend existing

programmes to meet workplace requirements and the professional and discipline needs for practice based on evidence. Qualifications can be described in terms of credits which directly reflect the required input of the students. In developing an education programme for radiation therapists, the qualification framework of the region needs to be applied.

2.8. EDUCATION ENVIRONMENT

The importance of providing an adequate learning environment for the students cannot be overemphasized. Both the physical infrastructure and the teaching staff need to be adequate. If they are not already in place, institutions should strive to attain the levels recommended below over a relatively short period of time. Even where resources are limited, a designated room is needed where student teaching and learning can take place.

2.9. PHYSICAL INFRASTRUCTURE

Teaching areas have to facilitate different teaching methods. Where students share didactic lectures with other disciplines, large lecture theatres may be appropriate, but smaller teaching areas need to also be provided for tutorial and problem/case-based learning approaches. In all venues where students are placed, health and safety standards have to be adhered to.

Computers with Internet access are essential for lecturers to maintain current knowledge, update information and communicate easily with national and international colleagues. Where space permits, a shared student room needs to be available.

A library with a sufficient variety of relevant textbooks, reference books and current journals needs to be available for staff and students. Appropriate resources may come in electronic form, namely open-access and subscription-based journals and textbooks. Where materials are solely electronic, equitable use of appropriate IT infrastructure (i.e. computers with Internet connectivity) has to be obtained by all students, either at their educational institute or their place of study. The appropriate computerized databases also need to be available. These facilities could be shared with other faculties in the educational institute. The IAEA has been providing educational materials on radiation oncology in general [13–19] and radiation therapy in particular to its Member States and can respond to appropriate requests through its Regular Programme and/or Technical Cooperation Programme.

2.10. ACADEMIC STAFF

Given the immense problems faced by many radiation therapy departments in low- and middle-income countries, there was agreement among members of the expert group that designating one person as the programme director would meet the short-term requirements. However, there has been concern around the advisability of establishing a training programme dependent on a single person. Consideration needs, therefore, to be given to funding one individual initially, followed as quickly as possible by a second lecturer to ensure sustainability. The ratio of lecturers to students has to be no more than 10 students per lecturer. To share the responsibility and decrease the single dependency situation there needs to be additional formalized radiation therapist positions. These have to be clinicians experienced in training and assessment contributing to the programme.

The minimum requirement for academic and clinical staff to be appointed has to be, at the very least, at a level higher than those taking the radiation therapy programme, for example at the master's degree level. This qualification can be gained by the radiation therapist programme

director while in post, if necessary. However, consideration has been given to a reasonable educational standard for practice tutors and educators in the radiation therapy clinical environment, whose qualifications pre-date this publication. A practice tutor is expected to be competent in all radiation therapy techniques and be able to educate and assess students.

2.11. PROGRAMME DIRECTOR/HEAD OF PROGRAMME

For a radiation therapy education programme, either stand-alone or as part of a combined programme, a programme director who is an experienced radiation therapist has to be appointed. Depending on the structure of the educational institution, the programme director/head of programme needs to have the appropriate level of authority and autonomy to make decisions that directly affect the day-to-day operations and development of the programme. They are responsible for the overall management of the programme and for ensuring that the criteria for the programme are implemented, requirements are met and that there is on-going development. It is also important that a programme development working group is put in place to support the director in developing the programme appropriately.

Where the education programme is university or higher education institution based the working group acknowledges that the first programme director may not have the academic qualifications required within the system. It is, however, essential that interim exceptions to the rule are made initially to facilitate the integration of the profession within the university setting led by an appropriately experienced radiation therapist. Provision needs to be made to enable this director to acquire higher academic qualifications over time.

The long-term aim should be to recruit sufficiently dedicated academic staff to meet national or institutional student/staff ratios as mandated. The IAEA focuses on ‘training the trainers’, which is crucial to the development of future education programmes and for maintaining an academic workforce. The dedicated staff have to maintain their own knowledge of the advances in radiation oncology, as well as current thinking on education philosophy and developments. Invited lecturers could assist in the initial programmes where dedicated staff numbers are limited.

It is desirable that the academic staff actively participate in national and international professional organizations, maintain close contact with the clinical staff and maintain their own knowledge and expertise in a rapidly developing profession. The development of national and international networks can assist in this.

2.12. PROGRAMME ACCREDITATION

Within the university or higher education institution an on-going accreditation system has to be in place with responsibility for monitoring the standards of the programme.

2.13. NATIONAL RECOGNITION

The recommendation is that the programme director and their team identify the responsible national education and regulatory authorities. Recognition of qualifications and professions varies between countries. It is highly likely that the Ministries of Health and Education will also have some jurisdiction in relation to professional standards and recognition and these have to also be consulted. In addition, there are international organizations that make recommendations relating to recognition which could impact on the development of the

programme. These organizations include the International Labor Organization (ILO) and the World Health Organization (WHO).

The International Basic Safety Standards (INTERNATIONAL BASIC SAFETY STANDARDS (BSS)) report [3] makes recommendations on radiation protection and safety for medical exposure. Regulatory requirements demand that health professionals with responsibilities for medical exposure are specialized in the appropriate area and that they meet the requirements for education, training and competence in radiation protection in the relevant specialty (IAEA-GSR Part 3) [3].). The programme director has to confirm the legal status of the profession within their own country. This needs to include whether statutory (state) registration is in place or whether recognition is delegated to a professional body. It is also useful to identify the situation in other countries with respect to national or international recognition of professional education programmes.

Policies with respect to freedom of movement of personnel within a region or between countries can also be helpful in determining the level of the national education programme.

2.14. PROFESSIONAL REPRESENTATION FOR RADIATION THERAPISTS

A profession is strengthened by the support of a dedicated professional body (association or society). This body can take the initiative in negotiations at government level to support the educational development in radiation therapy, including moving from diploma to degree level. It is important to carefully consider the various options for professional representation to ensure that maximum benefit is gained.

As with education, the optimum situation is a professional body that represents radiation therapists specifically. However, in some instances this will not exist and the team has to find out what the local situation is with respect to establishing professional bodies.

Collaboration with national societies representing radiation oncologists and medical physics experts is often very beneficial for the radiation therapist. In addition, international professional society membership have to be promoted and linkages made wherever possible.

3. TEACHING AND LEARNING METHODS

Learning is defined as “a process that leads to change, which occurs as a result of experience and increases the potential for improved performance and future learning” [19]. The change in the learner may happen at the level of knowledge, attitude or behaviour. At the first level of Bloom’s taxonomy, learning indicates an increase in the quantity of information that a student acquires and retains and acquisition of new facts or skills. At the higher levels of Bloom’s taxonomy, learning involves the processing of information acquired to make sense of it, abstracting meaning and identifying ways in which it can be applied to other situations. It thus becomes an active, interpretive process requiring higher level skills. This higher level of learning is necessary for radiation therapists because they work in a technological, dynamic and interactive discipline, where application of knowledge in decision-making is paramount to safe treatment preparation and delivery.

Students learn in different ways. Therefore, a wide range of learning experiences and assessment methods as is feasible have to be used to facilitate this spectrum of student learning styles. Teaching and learning necessitate that the lecturer be a facilitator of education, encouraging students to be more proactive in their learning and to seek knowledge for themselves; not being a provider of facts and figures. Students need to be taught how to find relevant information, how to evaluate it and how to organize it into a meaningful outcome for their practice.

Teaching is about motivating students to learn and how to learn in a manner that is relevant, meaningful and memorable. It is important for lecturers to remain abreast of current and upcoming developments within radiation therapy and cancer management and to be able to bridge the gap between theory and practice. Good teaching is about caring, nurturing and developing minds and talents. In radiation therapy it is particularly important that the clinical and academic components are integrated and reflective of each other.

It is a useful exercise to consider the teaching methods available in individual contexts and to list the strengths, weaknesses, limitations and level of preparation necessary for each, to evaluate the most appropriate and realistic method to meet the roles of this proposed framework.

3.1. TEACHING AND LEARNING METHODS

A wide range of teaching and learning methods are possible and the method has to reflect the audience, aims and learning outcomes. Resources frequently dictate the type of methods used and in this case the faculty has to try to optimize the method as far as possible within these constraints. Compromises may be necessary depending on the available resources, but the aim should be to move towards the optimal teaching method to achieve the key and enabling competences.

Below are short outlines of some commonly used teaching and learning methods that may be useful to consider but are not exhaustive.

3.2. LECTURE

This is the most traditional teaching method and has advantages as well as limitations. It is most suitable for large groups where the delivery of factual information in a didactic format is required. The greatest limitation is that the audience is often passive and communication is frequently one way. It is difficult in this context to gauge what students are learning outside of

the examination process. Where possible the lecturer has to give examples in practice of the topic they are teaching, break the lecture into discrete time blocks separated by a question, short discussion, worksheet to complete or some similar approach. This helps to maintain the student's concentration. This method is useful for example, when teaching anatomical principles, but needs to be followed up with practical application.

3.3. COMPUTER BASED

With access to a computer laboratory, it may be possible to use interactive software for teaching a small group of students, topics such as treatment planning and treatment verification. The benefit of such an approach is that it is closely aligned with clinical practice and affords students the opportunity to consider 'real life' cases, safe in the knowledge that mistakes can be made in this simulated environment. These can be the focus of discussion not only on treatment planning and verification but on incident and near miss reporting. Online learning also offers the potential for students to benefit from synchronous learning from guest lecturers outside the jurisdiction. Use of a student learning system permits asynchronous learning where lecturers can provide notes and readings for students to access in their own time.

3.4. PRACTICALS

Where the topic has a strong clinical element, such as positioning and immobilization, several approaches can be used. It may be possible to obtain immobilization accessory equipment that can be used for demonstration purposes. This may require some financial outlay. It may also be possible to use the radiation therapy clinical site outside of regular treatment times to demonstrate accurate positioning procedures and equipment manipulation.

3.5. TUTORIALS

Tutorials are an excellent way of encouraging student participation. They may be stand alone or offered to support lectures. In these tutorials the student is usually given some preliminary work to do and the tutorial then becomes an active discussion rather than a delivery of information. It is useful for clarification of problems or to stimulate students to take greater responsibility for their own learning. It is important to clearly identify the topic for discussion and to give references where material can be sourced.

3.6. JOURNAL CLUB

A journal club is similar to the tutorial, but the students are given a selection of journal articles on a specific topic and asked to analyze them and provide feedback during the session. This helps students to read articles critically and to differentiate between the levels of quality of the research carried out. This can be very useful in providing students with the most current information on a topic and enhances their critical thinking skills.

3.7. CASE STUDIES

Case studies are stories that are used as a teaching tool to show the application of a theory or concept to real situations. Depending on the goal they are meant to fulfil, cases can be fact-driven and deductive where there is a correct answer, or they can be context driven where multiple solutions are possible. Case studies provide students with scenarios in which they can begin to think about their understanding and solutions to problems found in real-world

situations. When carefully planned, case studies will challenge students' critical thinking and problem-solving skills in a safe and open learning environment.

Case studies serve as educational narratives that illustrate the application of theories or concepts in real-world contexts. Depending on their intended purpose, they can be fact-based and deductive, offering a single correct answer, or context-based, allowing for multiple possible solutions [20]. These scenarios encourage students to explore their understanding and develop solutions to real-life problems. When thoughtfully designed, case studies enhance students' critical thinking and problem-solving skills within a supportive learning environment.

This method can be very useful in radiation therapy as it directly relates theory and practice. Real clinical issues can be raised and discussed, and this helps the students to consider problems and ways in which they can be addressed. It allows individual experiences and situations to be considered and does not necessarily impose solutions that may not be applicable in every specific situation. This type of approach also encourages reflection on professionalism and is dependent on good material selection and skill by the lecturer.

3.8. PROBLEM BASED LEARNING

First developed in McMaster University, Canada [21] problem-based learning (PBL) is a student-centered approach where a problem is set to a group of students who have to divide the problem into separate components and then allocate tasks to individuals within the group. The group arranges meetings and feedback sessions where the information obtained is shared with the full group and discussed in the context of the problem. It is an excellent learning method but is resource intensive and often difficult to implement as a complete curriculum design where resources are limited. A modified scenario/PBL methodology can be implemented to achieve the advantages of PBL that encourages active learning and enquiry on the part of the student, promotes collaboration within the group as well as communication skills and provides the student with lifelong skills in terms of addressing difficulties that subsequently arise in the clinical setting.

3.9. SIMULATIONS/LABORATORY EXERCISES

Medical simulations, in general, aim to imitate real patients, anatomic regions or clinical tasks and/or to mirror the real-life circumstances in which medical services are rendered.

Virtual reality (VR) simulations are even newer innovations in which a computer display simulates the physical world and user interactions are with the computer within that simulated (virtual) world [22]. The learner is required to react to the simulation as they would under real-life circumstances; although it is understood that the fidelity of a simulation is never completely identical to the real world. Some reasons are obvious: engineering limitations, psychometric requirements, cost and time constraints. Nonetheless, technological advancement leading to higher fidelity and increasingly realistic simulations have been significant contributors to the recent rise in the use of this technology throughout medical education.

Virtual reality tools for the education of radiation therapists can simulate a virtual environment of a radiation therapy treatment room including a life size linear accelerator which is virtually controlled by the trainee using a real control handset. In this way students can have unlimited practice without using valuable accelerator time and without involving patients or equipment while enhancing the understanding of radiation therapy concepts and skills which are often difficult to teach in a classroom.

Simulation in medical education continues to grow [23]. There are advantages in terms of patient safety, reduction of errors and lower healthcare costs, due primarily to the enhancement of students critical thinking and psychomotor skills. There are some practical limitations to implementing simulation training programmes. These are faculty/academic staff time constraints, lack of educator training and cost of the equipment since many of the technologies implemented for simulated teaching and learning are expensive. Simulations for training medical students and radiation therapists cannot replace the direct contact and care of patients in the clinical environment. It is seen as a preparatory phase for practical clinical work. Best practice in the use of simulation in teaching includes briefing the student's pre-simulation on the exercise, debriefing after the exercise, coupled with the opportunity for self-reflection on performance.

4. ASSESSMENT

Assessment is the process of documenting, usually in measurable terms, the extent to which the learning outcomes have been achieved and can cover knowledge, skills and attitudes.

Assessment plays a very important part in the education process and often dictates what a student will learn. It should not simply be about the allocation of grades but needs to help inform and support student progress and identify areas where additional input is required. Assessment has to be seen as facilitating learning and needs to focus on what is learnt rather than what is taught consistent with the focus on learning outcomes.

Assessment should be transparent and authentic to reflect the future work of students upon graduation. Assessment should assist rather than intimidate students. In this context assessment should reflect the learning outcomes and measure to what extent they have been met, which then is an evaluation of the effectiveness of the teaching process. It can be used by the faculty to measure how effective the linkages between the learning outcomes and the teaching methodology have been and indicate areas where further review is required.

Assessment can be classified in many ways, some of the most usual are:

- Formative or summative;
- Objective or subjective;
- Formal or informal.

Formative assessment takes place throughout a module or project and is used to aid learning and give continuous feedback on performance to students and motivate their learning while *summative assessment* occurs at the end of a programme or module, and its purpose is generally to enable the awarding of a grade. These two methods are routinely used in larger modules to complement one another.

Objective assessment, in its simplest explanation, is the use of a form of questioning where there is a single correct answer. This could be something like a multiple-choice question (MCQ) or a mathematical calculation as in dose/fractionation calculations. *Subjective* on the other hand may have more than one correct answer or there may be more than one way of answering the question. Essays can be considered in this way and again an example would be the treatment of a tumour site where more than one option could be considered as correct.

Informal assessment does not usually require a written answer and can be very useful in guiding students during class or practical sessions. Informal assessment can include observation, peer and self-evaluation, discussion or checklists. *Formal assessment*, on the other hand, usually implies a written examination/assignment.

Assessment is one of the most obvious ways to evaluate what the student has understood whether they can apply the knowledge and/or carry out the practical skill or whether they have developed affective skills such as good communication. Assessments that are authentic to the workplace can inspire learning by affording the student the opportunity to apply their knowledge and skills. Assessment is also a means of evaluation of the effectiveness of the programme as a whole and its individual components.

Assessments are an integral component of programme design, and the amount and level of assessment has to be consistent with the defined learning outcomes. It is about finding out what the student has achieved and giving it value.

There needs to be an assessment of student performance in relation to both the theoretical and clinical components of the programme and these have to be clearly stated [8].

A programme developed from this IAEA framework can employ a wide range of assessment tools which can be administered throughout the programme as appropriate to the stage of education of the students. Methods of assessment will vary across the various countries or regions.

Assessment may be carried out by the lecturers, clinical staff, peer groups and the learners themselves. In addition to individual assignments, progressive assessment of knowledge, understanding and skills of the students have to be made throughout the duration of the programme. Evaluation is designed to monitor the progress of a programme and the degree to which the objectives it is intended to meet are being implemented.

Clearly identifiable objectives need to be set for evaluation during programme planning. These may relate to the theoretical content of the programme, the practical or clinical setting, the teaching staff and the available resources. All aspects of programme design and planning may vary depending upon the environment in each country, but each will need to be considered by the team developing the curriculum. The programme planning team will be involved in monitoring the quality of the programme and its outcome and in making revisions as necessary.

4.1. ASSESSMENT METHODS

Like teaching and learning methods, assessment can take many forms thereby testing a wide range of knowledge, skills and attitudes consistent with the taxonomy defined by Bloom [10, 24, 25]. Teaching, learning and assessment methods have to align. For example, it is not optimal to teach a practical skill to students using computer-based technology and then to examine it in a written examination. Similarly, assessing student communication skills using an essay will not indicate to the lecturer their ability to provide information and support to patients.

In all assessments that will be allocated a mark or grade, it has to be made clear to the student as to how the marks are going to be allocated in a rubric. This will also indicate to them the level of detail required on each aspect of the topic, known as the assessment criteria. A short overview of some of the more commonly used assessment methods is given below.

4.2. TRADITIONAL EXAMINATION

Traditional examinations are usually held at the end of a block of learning, either a module or the end of a full academic year. The method ensures that what is returned is the student's own work. They do not, however, usually allow for feedback and learning as they take place at the end of a module/programme.

It is noted that some students are naturally better at sitting examinations, have greater ability to recall large numbers of facts, to set out answers clearly and even to write faster than others. The answers may, however, not always reflect either the level of understanding of the student or their ability to apply knowledge in a wider context. Since it is quite challenging to write clear and unambiguous examination questions it is useful to write them with at least one other lecturer

and to ask another to read them to ensure clarity. The examiner has to be very clear on what the focus of the question is and what exactly is being tested. It is useful for the examiner to answer the question personally and to use this as a guide to marking. However, the examiner needs to remember that there may be other points of view or approaches to answering the question and one need not penalize students who do not give back exactly what was drafted in the sample answer. This necessitates a high level of knowledge on the part of the examiner.

4.3. OPEN BOOK EXAMINATION

This is like the traditional examination, but the student can take source or reference material into the examination room. This is a useful method when a lot of factual information that does not necessarily have to be learnt and remembered by the student, is required in the answer. This method of assessment can also be used to assess how well students can use sources or reference material and apply the information gained to a specific question, which indicates their ability to problem solve.

Types of questions used

Essay type questions

Long essay type questions are used where students are expected to apply principles or knowledge to a given situation. Students will usually be expected to give an opinion or to draw from other experiences they may have had. This type of question gives the student the opportunity for self-expression and to show wider reading or understanding and has to be marked with this in mind. Long essay questions can disadvantage students whose vocabulary is limited or who have poor writing skills.

Short answer questions

Short answer questions are used to test a single more specific aspect of a topic. They are used to test a breadth of knowledge over a wider range of topics. The examiner may prepare an entire examination paper of short answer questions or may choose to include one or longer essay questions. What is important is that all the questions reflect the learning outcomes defined initially in the module or unit of learning.

Multiple choice questions examinations

In this form of assessment students are asked a question or presented with a statement, the 'stem' and given a set of possible answers from which, in its simplest form, they are required to select the correct answer. The correct answer is termed the 'key' and the incorrect answers are the 'distractors'. Stems have to be clear; with the key leaving no area for ambiguity (it needs to be well referenced) and the distractors should be incorrect but minorly so. They need to test the students' knowledge directly. MCQs can take a range of forms and can include diagrams, short scenarios or case studies. It is also possible to test depth of knowledge by developing a linking series of MCQs on a single topic. MCQs can be a very objective method of assessment.

4.4. CLINICAL PORTFOLIO

The clinical portfolio is a compilation of a range of work built up over a period and has the advantage of showing evidence of the student's achievement over time. One disadvantage is for the lecturer as they are time-consuming and can be subjective to mark.

4.5. REFLECTIVE DIARIES

These allow the students to record their experiences and to reflect on what they have learnt. They help students to focus on what they have learnt and how they have integrated this into their own thinking about how they would approach a similar scenario again. They are used to record experience and to help understanding, to increase the ability to consider a problem or experience and reflect on how the student may address it. It is applicable in the clinical setting to encourage practitioners to think about what they are doing, why they are doing it and whether there is a way to do it better.

4.6. OBJECTIVE STRUCTURED CLINICAL EXAMINATION

These are frequently used in the medical setting to test practical skills or affective skills such as communication [26] and specifically for practical topics that cannot easily be assessed using other means. They normally consist of several stations, where a station is a discrete section within a specially laid out room where a single topic is assessed. Students are expected to move through each of the stations and to answer the problem set within a defined period (usually 5–10 minutes).

The stations may ask a dose calculation question or may set a case scenario and ask the student how they would deal with it. In more sophisticated systems, actors may be used to present the student with a problem where their ability to communicate can be tested. OSCEs are very useful in testing a range of skills but rely upon assessment criteria that are specific, determined in advance and documented. Great care has to be taken in both setting up and marking the stations.

4.7. PRESENTATION (INDIVIDUAL OR GROUP)

This can be a very good method of assessment as it demonstrates the depth of understanding on a given topic as well as the ability to collect and filter evidence used in the presentation to support the points made. Group presentations have the added benefit of assessing team dynamics and communication skills.

4.8. POSTERS

Posters are useful in requiring students to present evidence in a concise fashion. They have to be supplemented with a discussion session to ascertain the depth of knowledge and critique of the student on the topic presented.

4.9. FEEDBACK POST ASSESSMENT

Feedback is an essential component of teaching, learning and assessment processes. Students cannot be expected to improve their performance without continuous, constructive feedback. Feedback needs to focus on what the student has done well as well as providing constructive comments on how they can improve their performance in the future on their areas for improvement.

5. PRACTICE EDUCATION

5.1. PRACTICE EDUCATION

Practice education is central to the development of the key competences associated with the seven radiation therapist roles (Radiation Therapy Expert, Communicator, Collaborator, Leader and Advocate, Quality Care Provider, Evidence Based Practitioner and Professional). The student will begin to understand how the knowledge and skills gained during the academic year(s) apply to clinical practice and will be supported in this by working closely with the clinical educators.

As part of placement the student is expected to familiarize themselves with the hospital and the radiation therapy department in particular, the staff involved in the management of the patient, how the team interacts with each other, with the wider multidisciplinary team and with the patient.

Practice educators (radiation therapists in the clinical department) are expected to support the students in gaining understanding and in developing their technical and psychosocial skills. The clinical environment is essential in providing students with the practical learning experience underpinning their future practice. The on-going engagement of clinical radiation therapists is essential for an effective, high-quality programme.

In the clinical environment the students learn to put their theoretical knowledge into practice and, at senior level, to critically analyze their practice. Radiation therapists have a responsibility to support the education and training of students, as their future colleagues.

5.2. CLINICAL PLACEMENT

Students need to be exposed to the widest possible clinical experience. This may not be possible where radiation therapy infrastructure is limited and necessitates clinical placement outside the locality or even the country. As a requirement, students have to spend significant time on treatment units. The student has to be placed where they can gain experience with a range of imaging modalities for verification and CT scanners/simulators for treatment planning purposes. It is also recommended that students have exposure to brachytherapy and other techniques and technologies, as appropriate to the context. The students have to learn how to evaluate a standard treatment plan in the educational environment but where need to also spend time in the treatment planning facility or be taught these skills in the educational institute, should resources permit. Students need to also attend QA, radiation safety, clinics and clinical meetings where these are relevant.

Clinical placements have to be structured to avoid overloading the clinical department. The number of placement areas need to be carefully considered, and students scheduled optimally to integrate into the clinical care of patients and operations.

5.3. CLINICAL EDUCATORS

Clinical education is a partnership between the educational institution and clinical department and may necessitate a formal agreement. The overall responsibility of the management of the clinical education programme lies with the educational institute. A member of the educational institute staff needs to be responsible for overseeing the clinical education programme. This has to be a radiation therapist with significant experience in both clinical practice and education

and is known as the practice education coordinator (PEC). They oversee the entire clinical practice programme and report to the programme director. The PEC liaises closely with the practice tutors, who are clinical radiation therapists in individual departments with responsibility for the management and education of students. These, in turn, work closely with all clinical radiation therapists who will take responsibility for the student's progress in the clinical setting in their practice education role.

5.4. PROFESSIONAL DEVELOPMENT

During clinical placement, the student will develop as a healthcare professional and gain an understanding of:

- How the radiation therapy department relates to and interacts with the other departments in the clinical environment;
- The patient pathway and the personnel involved at each stage;
- The radiation therapy team and the wider multidisciplinary team;
- Their role and the appropriate communication skills and professional attitude and conduct in the clinical setting.

By the end of the senior level of practice of education assessment, the student needs to have achieved all outlined competencies at entry-level proficiency.

5.5. CLINICAL ASSESSMENT

Clinical assessment is an essential component of the clinical programme. It is the method by which the student's competence in the practical setting is evaluated. To maintain objectivity and standardization in the assessment process, it is essential to use an assessment tool that aligns with the expected competencies and teaching methods.

The clinical assessment tool monitors the student's progress and tests increasing levels of knowledge, skills, application and judgement at each placement and the consistency with which it is performed. Assessments have to be both formative and summative and need to include a statement of an observable performance by the student and an indication of the minimum acceptable performance standard. Assessment tests the quality and efficiency with which the student demonstrates a specific competence. Students have to complete a series of assessments during their clinical programme. Examples of assessment tools that can be used for pre-treatment and treatment delivery are given in Appendix I, together with the expected levels across domains at junior student and senior student level. '**Junior student**' refers to students completing initial practice placements at the start of their practice education programme. '**Senior student**' refers to students completing all subsequent placements. These can therefore be adopted into all contexts and programme durations, as appropriate, however, all students have to be competent at the '**Senior student**' level in practice education in both pre-treatment and treatment delivery before they can be deemed eligible to graduate from any radiation therapy programme.

To facilitate knowledge transfer from theory to practice, students have to complete case reports covering a range of routine disease sites observed in the region. Students need to build a clinical portfolio including practical assessments and case reports across the duration of their programme.

Evaluation of the clinical placement by the student is also important and will allow the educational institute and clinical department staff to consider the standard of clinical teaching and the teaching support and resources. Radiation therapists need to adequately teach and assess students. It also encourages the students to take more control of their own educational experience with the opportunity to monitor their own progress and to give feedback to the clinical departments.

5.6. FEEDBACK

It is important to allow the student to give feedback and comment on their experience of each practice placement. This allows them to reflect on their learning and to consider what their focus will be on the next placement block. Placement supervisors have to receive feedback on their assessment of student progress. Feedback between placement supervisors and the programme director needs to be two-way to ensure synergy between the academic component and clinical practice

APPENDIX

The Practice Education Assessment Forms (Pre-Treatment and Post Treatment) serve as important tools for evaluating a learner's readiness to participate safely and effectively in radiation therapy treatment delivery. Below are examples of these forms with sections containing personal data, expected competencies, assessment of professionalism, patient management skills and a rubric for performance assessment.

A.1. PRACTICE EDUCATION ASSESSMENT FORM (PRE-TREATMENT)

TABLE 2. PERSONAL DATA OF TRAINEE

Name of Pre-Treatment Area		
Radiation Therapy Department		
Placement Dates		
Attendance record	Detail	Comment
Full attendance		
No. of days absent		
Dates absent		

Expected competencies at junior student level include:

- (a) Always adhering to departmental health and safety procedures, including radiation protection and infection control guidelines.
- (b) Contributing to the set-up of patients on the treatment unit under the supervision of qualified radiation therapist:
 - (i) Preparing the treatment room and any immobilization devices;
 - (ii) Identifying patients according to local procedure;
 - (iii) Checking patient condition and ensure patient has completed all necessary pre-CT/simulation preparation measures;
 - (iv) Positioning and immobilizing the patient under the supervision of qualified radiation therapist;
 - (v) Building rapport with radiation therapy team and patients, demonstrating good communication skills;
 - (vi) Confirming scanning parameters/simulation field borders;
 - (vii) Being professional in appearance and conduct at all times.

Expected competencies at senior student level include:

- (a) Always adhering to departmental health and safety procedures, including radiation protection and infection control guidelines.
- (b) Being competent in all treatment room procedures:
 - (i) Completing pre-treatment preparation procedures for patients for any of the most observed cancer sites in the radiation therapy department;
 - (ii) Problem solving on set as required;
 - (iii) Appropriately communicating with patients at all stages of pre-treatment procedures;
 - (iv) Participating in all documentation procedures;
 - (v) Participating as a full team member;
 - (vi) Being professional in appearance and conduct.

TABLE 3. PROFESSIONALISM

Student Skills	Pass	Fail
Professional Appearance		
Attendance		
Punctuality		
Courtesy, Attitude and Respect		
Reliability		
Respected protection laws and patient confidentiality		
Clear understanding and application of AS LOW AS REASONABLY ACHIEVABLE (ALARA) principles		
Demonstrated good hand hygiene practices		
Demonstrated knowledge of health and safety procedures (e.g. Fire, infection and emergency procedures and location of all emergency equipment)		
OVERALL COMMENTS:		

TABLE 4. PATIENT MANAGEMENT

The student:	Unacceptable	At expected standard	Above expected standard	Well above expected standard	Exceptional
Identified patients as per the local procedure					
Communicated effectively with patients and carers					
Assisted and supported patients throughout treatment procedures					
Evaluated patient condition					
Understood risks with respect to cannulation, use of contrast and tattooing and acted appropriately					
Followed good hygiene practices and infection control policies					
Communicated effectively and worked as a team member in positioning patients					
Demonstrated clear rationale in the selection of immobilization devices					
Participated in immobilization device construction, as appropriate					
Identified region to be scanned and determined appropriate scanning/field borders					
Adhered to AS LOW AS REASONABLY ACHIEVABLE (ALARA) principle at all times during border/field selection and scanning/fluoroscopy					

TABLE 4. PATIENT MANAGEMENT (CONT.)

The student:	Unacceptable	At expected standard	Above expected standard	Well above expected standard	Exceptional
Supported patient throughout the pre-treatment procedures					
Communicated effectively with patients and carers at all times					
Documented all pre-treatment parameters					
OVERALL COMMENTS:					

A.2. RUBRIC FOR PERFORMANCE ASSESSMENT

Table 5 provides an example of a rubric for performance assessment. This rubric outlines to the student and the clinical assessor the criteria to be met by the student at junior and senior level. A student is required be competent at the 'senior level' before they can be considered for graduation.

TABLE 5. RUBRIC FOR PERFORMANCE ASSESSMENT

Criterion	Junior Level	Senior Level
Radiation therapy knowledge	Knows rationale for the likely RT technique and how the position selected at CT/simulation will affect this.	Demonstrates a breadth of knowledge of radiation therapy treatment concepts and patient management and can critically analyse practice.
Degree of independence	Dependence on radiation therapist instruction for scanning parameters but able to select correct immobilize device and position appropriately. Able to check patient condition.	Rarely requires direction or prompting during pre-treatment procedures for the main cancer sites observed. Able to appropriately select patient position, immobilize and determine scanning parameters/field borders.
Time efficiency	Regularly requires additional task time on the treatment unit, although this lessens towards the end of the placement block.	Time efficient for the pre-treatment preparation of the main cancer sites observed.
Interpersonal skills	Effective contributor to the radiation therapy team and developing skills within the wider multidisciplinary team. Professional at all times. Communicates with patients, carers and all staff.	Demonstrates own initiative and has excellent teamwork skills. Advanced communication skills across all settings.

A.3. PRACTICE EDUCATION ASSESSMENT FORM (TREATMENT DELIVERY)

TABLE 6. PERSONAL DATA OF TRAINEE

Name of Treatment Unit		
Radiation Therapy Department		
Placement dates		
Attendance record	Detail	Comment
Full attendance		
No. of days absent		
Dates absent		

Expected competencies at junior student level include:

- (a) Always adhering to departmental health and safety procedures, including radiation protection and infection control guidelines.
- (b) Contributing to the set-up of patients on the treatment unit under the supervision of qualified radiation therapist:
 - (i) Preparing the treatment room and any immobilization devices;
 - (ii) Accurately identifying patients according to local procedure;
 - (iii) Checking patient condition and ensure patient has completed all necessary pre-treatment preparation measures;
 - (iv) Positioning and immobilizing the patient under the supervision of qualified radiation therapist;
 - (v) Building rapport with radiation therapy team and patients, demonstrating good communication skills;
 - (vi) Confirming treatment parameters;
 - (vii) Being professional in appearance and conduct.

Expected competencies at senior student level include:

- (a) Always adhering to departmental health and safety procedures, including radiation protection and infection control guidelines.
- (b) Demonstrating competency in all treatment room procedures:
 - (i) Being able to set up patients for any of the most observed cancer sites in the radiation therapy department, including problem solving as required;
 - (ii) Appropriately advising on and managing patient acute side effects;
 - (iii) Reviewing on treatment images and adhering to imaging protocols under the guidance of qualified radiation therapist;
 - (iv) Providing patient information from the first to the final day of treatment;
 - (v) Participating as a full team member;
 - (vi) Being professional in appearance and conduct.

TABLE 7. PROFESSIONALISM

Student Skills	Pass	Fail
Professional Appearance		
Attendance		
Punctuality		
Courtesy, Attitude and Respect		
Reliability		
Respected protection laws and patient confidentiality		
Demonstrated good hand hygiene practices		
Demonstrated knowledge of health and safety procedures (e.g. Fire, infection and emergency procedures and location of all emergency equipment)		
OVERALL COMMENTS:		

TABLE 8. PATIENT MANAGEMENT

The student:	Unacceptable	At expected standard	Above expected standard	Well above expected standard	Exceptional
Identified patients as per the local procedure					
Communicated effectively with patients and carers					
Assisted and supported patients throughout treatment procedures					
Evaluated patients' physical signs and symptoms and treatment side effects and was aware of possible psychosocial issues					
Discussed the appropriate management of side effects with patients and reported problems/queries to the RT team as appropriate (Senior)					
Followed good hygiene practices and infection control policies					
Communicated effectively and worked as a team member in positioning patients					
Positioned patients in a reasonable length of time (Senior)					
Positioned patients in an appropriate and safe manner					
Used the correct accessory devices on this unit					
Used the correct immobilization devices for each patient on this placement					

TABLE 8. PATIENT MANAGEMENT (CONT.)

The student:	Unacceptable	At expected standard	Above expected standard	Well above expected standard	Exceptional
Performed the final checks (Focus To Skin Distance (FSD), Bolus, Multi-Leaf Collimators (MLC)s...) and verbal confirmation of setup parameters in the treatment room (Senior)					
Participated in image verification procedures (Senior)					
Participated in vivo dose verification procedures, if applicable (Senior)					
OVERALL COMMENTS:					

TABLE 9. RUBRIC FOR PERFORMANCE ASSESSMENT

Criterion	Junior Level	Senior Level
Radiation knowledge therapy	Knows the rationale for the chosen RT technique as well as imaging, positioning and immobilisation and basic patient management.	Demonstrates a breadth of knowledge of radiation therapy treatment concepts and patient management and can critically analyse practice.
Degree of independence	Dependence on radiation therapist instruction for patient set-up. Able to check patient condition.	Rarely requires direction or prompting during daily set-up for the main cancer sites treated on the unit. Able to manage side effects for patients of all cancer sites on the unit.
Time efficiency	Regularly requires additional task time on the treatment unit, although this lessens towards the end of the placement block.	Considered time efficient for the treatment delivery of the main cancer sites observed.
Interpersonal skills	Effective contributor to the radiation therapy team and developing skills within the wider multidisciplinary team. Professional attitude at all times. Communicates with patients, carers and all staff.	Demonstrates own initiative and has excellent teamwork skills. Advanced communication skills across all settings.

TABLE 10. CASE REPORT TEMPLATE

Student identification:	
Radiation Therapy Department:	
Linac/Cobalt/CT/Simulator:	
Dates of practice education placement:	
Cancer site of this report:	

Q.1. Please give a summary of the patient history and other relevant information.

Age	
Gender	
Site	
Pathological classification	
Stage of disease	
Grade of disease	
radiation therapy Prescription	
Presenting signs and symptoms	
Patient risk factor(s), if appropriate	
Relevant medical and social history	
Investigations performed	
Performance status	
Treatment regimen	

Q.2. Describe and discuss

- The overall management plan for this patient
- The radiation therapy process from simulation/CT through treatment planning to treatment delivery
- The image and dose (if applicable) verification process

Q.3. Describe the acute and late side effects associated with radiation therapy to this site and outline how the acute side effects can be managed by you as a radiation therapist.**Q.4. Consider the psychosocial care required by this patient. Discuss your role in providing this support.**

Adapted from the clinical programme of the Discipline of Radiation Therapy, School of Medicine, Trinity College Dublin.

REFERENCES

- [1] INTERNATIONAL ATOMIC ENERGY AGENCY, Transition from 2D Radiotherapy to 3D Conformal and Intensity Modulated Radiotherapy, IAEA-TECDOC-1588, IAEA, Vienna (2008).
- [2] COFFEY, M., et al. A core curriculum for RTTs (radiation therapists/radiotherapy radiographers) designed for developing countries under the auspices of the International Atomic Energy Agency (IAEA). *Radiother. Oncol.* **81** (2006) 324–325. <https://doi.org/10.1016/j.radonc.2006.08.026>
- [3] INTERNATIONAL ATOMIC ENERGY AGENCY, Radiation Protection and Safety of Radiation Sources: International Basic Safety Standards. IAEA Safety Standards Series No. GSR Part 3, Vienna (2014) <https://doi.org/10.61092/iaea.u2pu-60vm>
- [4] BLOOM, B.S., (Ed), Taxonomy of Educational Objectives: The Classification of Educational Goals: Handbook I, Cognitive Domain. New York; Toronto: Longmans, Green (1956).
- [5] ANDERSON, L.W. KRATHWOHL, D.R. (Eds.) A Taxonomy for Learning, Teaching and Assessing: A revision of Bloom's Taxonomy of Educational Objectives: Complete Edition, New York: Longman (2001).
- [6] SHERBINO, J., FRANK, J.R., Educational Design. A Canadian Medical Education Directives for Specialists (CanMEDS) Guide for the Health Professions, Royal College of Physicians and Surgeons, Canada (2011).
- [7] ROYAL COLLEGE OF PHYSICIAN AND SURGEONS OF CANADA, The CanMEDSframework., <https://www.royalcollege.ca/en/standards-and-accreditation/canmeds>
- [8] ERIKSEN, J.G., et al., The updated ESTRO core curricula 2011 for clinicians, medical physicists and RTTs in radiotherapy/radiation oncology, *Radiother. Oncol.* **103** (2012) 103–108. <https://doi.org/10.1016/j.radonc.2012.02.007>
- [9] PÖTTER, R., et al., Competencies in radiation oncology: a new approach for education and training of professionals for radiotherapy and oncology in Europe. *Radiother. Oncol.* **103** (2012) 1–4. <https://doi.org/10.1016/j.radonc.2012.03.006>
- [10] COFFEY, M., et al. The European core curriculum for radiotherapy technologists. *Radiother. Oncol.* **43** 1 (1997) 97–101. [https://doi.org/10.1016/s0167-8140\(97\)01901-4](https://doi.org/10.1016/s0167-8140(97)01901-4)
- [11] COFFEY, M., DEGERFA, J., OSZTAVICS, A., VAN HEDEL, J., VANDEVELDE, G., Revised European core curriculum for RTs. *Radiother. Oncol.* **70** (2004) 137–158. <https://doi.org/10.1016/j.radonc.2003.12.001>
- [12] Bologna Working Group on Qualifications Frameworks. A Framework for Qualifications of the European Higher Education Area. Ministry of Science, Technology and Innovation. Copenhagen, The European Qualifications Framework, Denmark (2005). https://ehea.info/media.ehea.info/file/WG_Frameworks_qualification/71/0/050218_QF_EHE_A_580710.pdf
- [13] INTERNATIONAL ATOMIC ENERGY AGENCY, Commissioning and Quality Assurance of Computerized Treatment Planning Systems for Radiation Treatment of Cancer, IAEA, Vienna (2004).
- [14] INTERNATIONAL ATOMIC ENERGY AGENCY, Planning National Radiotherapy Services: A Practical Tool, IAEA, Vienna (2010).
- [15] INTERNATIONAL ATOMIC ENERGY AGENCY, Setting up a Radiotherapy Programme: Clinical, Medical Physics, Radiation Protection and Safety Aspects, IAEA, Vienna (2008).
- [16] INTERNATIONAL ATOMIC ENERGY AGENCY, Comprehensive Audits of Radiotherapy Practices: A Tool for Quality Improvement, IAEA, Vienna (2007).
- [17] INTERNATIONAL ATOMIC ENERGY AGENCY, Transition from 2D Radiotherapy to 3D Conformal and Intensity Modulated Radiotherapy, IAEA TECDOC-1588, Vienna (2008)

- [18] INTERNATIONAL ATOMIC ENERGY AGENCY, Management of Cervical Cancer: Strategies for Limited resource Centers – A Guide for Radiation Oncologists, IAEA, Human Health Reports 6, Vienna (2013).
- [19] AMBROSE, S.A., BRIDGES, M.W., DIPIETRO, M., LOVETT, M.C., AND NORMAN, M.K., How Learning Works: Seven Research-based Principles for Smart Teaching., San Francisco (2010).
- [20] MCLEAN, S.F., Case-Based Learning and its Application in Medical and Health-Care Fields: A Review of Worldwide Literature, J Med Educ Curric Dev. **27**(3) (2016). doi: 10.4137/JMECD.S20377
- [21] WOOD, D.F., Problem based learning, BMJ (2003); 326–328.
- [22] SCALESE, R.J., OBESO, V.T., ISSENBERG, S.B., Simulation technology for skills training and competency assessment in medical education, J. Gen. Intern. Med. **23** (1) (2007) 46–49.
- [23] OKUDA, Y., et al, The utility of simulation in medical education: what is the evidence? Mount. Sinai. J. Med. **76** (2009) 330–343.
- [24] ROSENBLATT, E., HAFFTY, B.G., LEER, J.W., Assessing Clinical Competence in Radiation Oncology Education, Radiology Education: the Evaluation and Assessment of Clinical Competence, In: (HIBBERT, K.M., CHHEM, R.K., VAN DEVEN, T., WANG, S., Eds.) Springer (2012) 117–128.
- [25] DAUPHINEE W.D., Assessing clinical performance: where do we stand and what might we expect? JAMA **274** (1995) 741–743.
- [26] SLOAN D.A., et al., Objective structured clinical examination (OSCE) for evaluation and instruction in graduate medical education, J. Surg. Res. **63** (1996) 225.

BIBLIOGRAPHY

American Society of Radiation Oncology (ASTRO) – <http://www.astro.org/>

American Society of Radiologic Technologists – <https://www.asrt.org/>

European Society of Radiotherapy and Oncology (ESTRO) – <http://www.estro.org/>

IAEA Human Health Campus – <http://nucleus.iaea.org/HHW/Home/index.html>

Telemedicine Networks: Asia-Pacific Radiation Oncology Network (ASPRONET), Africa

Radiation Oncology Network (AFRONET)

GLOSSARY

didactic teaching and learning: Learning where the teacher provides information and the students are relatively passive.

clinical/practice learning: Practical skills education and learning in the radiation therapy department.

Competency: Core ability.

key competency: The knowledge, skills and judgement required of a radiation therapist at entry to practice.

enabling competency: More detailed deliverable of the key competency. Describes the knowledge and skills that contribute to the successful outcomes or delivery of the key competency.

ABBREVIATIONS

ALARA	as low as reasonably achievable
IBSS	International Basic Safety Standards
CanMEDS	Canadian Medical Education Direction for Specialists
CT	computed tomography
FSD	focus to skin distance
ICRU	International Commission on Radiation Units and Measurements
ILO	International Labor Organization
IT	information technology
MCQ	multiple choice questions
MLC	multi-leaf collimator
OSCE	objective structured clinical examination
PBL	problem based learning
PEC	practice education coordinator
QA	quality assurance
RT	radiation therapy
VR	virtual reality
WHO	World Health Organization

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